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Original Article

Experimental investigations on mechanical properties of multi-layered structure fabricated by GMAW-based WAAM of SS316L

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ABSTRACT

In the present study, the Gas metal arc welding (GMAW) based Wire-arc additive manufacturing (WAAM) process was used to fabricate a multi-layered structure at optimized process parameters on SS316L using metal wire of SS316L. The multi-layered structure's microstructure, macrostructure, and mechanical properties (tensile test, impact test, microhardness, and fractography) were examined at three locations at the top, middle, and bottom sides of the structure. Macrostructure at different zones has confirmed an appropriate bonding between the two layers, complete fusion without oxidation, and free from defects and unwanted geometries. Microstructure results have observed a colony of columnar dendrites in the bottom zone, coarser grains with vertical growth along with the residual ferrite in the middle zone, and vertical dendritic structure with residual ferrite in skeletal shape in the top zone. Results of all tensile properties for top, middle and bottom zone developed by the WAAM process fall in the range values of wrought SS 316 L. The microhardness values were shown a consistent behavior across the built structure in all three zones. The obtained average value for the impact test has shown better strength than commercially used wrought SS 316 L. The results of fractured tensile and fracture impact test specimens revealed many dimples, which suggests a good ductility of the as-built structure. Thus, the obtained results have shown that the built structure using the GMAW-based WAAM process matches the standards for industrial applications.

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Nomenclature

AM	Additive manufacturing
CMT	Cold metal transfer
DED	Direct energy deposition
GMAW	Gas metal arc welding
MH	Microhardness
SEM	Scanning electron microscopy
SS	Stainless steel
TS	Travel speed
UTS	Ultimate tensile strength
WAAM	Wire arc additive manufacturing
WFS	Wire feed speed
YS	Yield strength

1. Introduction

The ability of additive manufacturing (AM) to directly manufacture a product by depositing material in one layer on top of another is gaining popularity. Compared to the traditional procedure, AM technique saves roughly 50% on fabrication costs [1,2]. As a result, it is regarded as a cost-effective, practicable, inventive, and dependable production technique [3]. The most significant benefit of AM technique is that it saves material waste and production time [4]. In recent years, advances in additive manufacturing technology have enabled the direct production of final parts. Its use is growing in various areas, including automotive, aerospace, defense, medicinal, and electronics, due to its numerous benefits [5–7]. AM is majorly divided into three classifications. Direct energy deposition (DED), sheet lamination, and powder bed fusion. Wire arc additive manufacturing (WAAM) with wire feedstock as the energy source was the DED process employed in AM technique [8,9]. WAAM can be obtained from various additional energy sources like plasma arc welding, gas tungsten arc welding, and gas metal arc welding (GMAW). WAAM-based technology using a GMAW source is employed for high deposition rate, favorable mechanical qualities, and low equipment cost, regardless of the typical GMAW process [10].

From the available techniques of AM, WAAM offers additional benefits. Because of its low initial cost, it is ideal for large-scale production. WAAM is a cost-effective technology since components such as a wire feed system, a welding torch, and an automated computer system are readily accessible [11]. Unlike the metal powder option, the metal wires necessary for WAAM are readily available and easy to handle [12]. High rate of deposition of around 8 kg/h can be achieved by using GMAW-based WAAM process. Even though high metal deposition limits surface quality and dimensional precision, it has the capacity to produce enormous volumes of components [13]. In order to reach the specified tolerances, post-processing like as milling or grinding may be necessary. Distortion and strains gets formed due to fast deposition rate, high temperature, and excessive energy input. These difficulties are usually associated with additive manufacturing processes. Path planning should be studied to acquire favorable process parameters such as trip speed, fee rate, and arc voltage [14].

316 L stainless steel (SS) has good corrosion resistance and mechanical qualities owing to the inclusion of elements like Mo, Ni, and Cr [15]. In view of good weldability, the SS 316 L has a higher welding temperature between 1300°C-1400 °C, which are excellent for the fusion between two layers in AM [16,17]. SS 316 L wires are primarily used in GMAW because of the higher deposition rate [18]. Due to the higher deposition rate, high-strength components can be easily made with good mechanical properties. SS 316 L is a low-carbon austenitic stainless steel with a carbon content of 0.03 percent by weight. Due to its excellent corrosion resistance, outstanding weldability, high strength and ductility, robust biocompatibility, and low cost, SS 316 L is the preferred material for industrial applications of maritime, offshore, biomedical equipment, cars, petrochemical facilities, and nuclear reactors [18].

Jin et al. [19] investigated the recent inclination towards the WAAM process for stainless steel alloys. Their study included investigating WAAM variables and their effect on changes in microstructure, mechanical properties, and other defects related to the fabricated multi-layered structure. Their extensive literatute on metallurgical aspects of WAAM specimens revealed that for the fabrication of good superior parts of stainless steels throught WAAM, it is necessary to analyze the physical metallurgy mechanisms of WAAM and necessity of post heat treatments to optimize the WAAM variables.

Chen et al. [20] studied the SS316L additive manufacturing with GMAW process. The research's major goal was to determine the impact of heat treatment on tensile strength and corrosion behavior. The findings revealed that increased temperature and period of heat treatment improved the corrosion properties of SS 316. Their findings revealed that it was due to the emergence of a complete austenitic phase in the matrix, which promoted corrosion. The ultimate tensile strength and yield strength were enhanced by heat treatment carried out at 1000 °C. Also, the maximum strength of the AM components was found to be equivalent to that of wrought 316 L. Queguineur et al. [12] explored WAAM variables intending to enhance mechanical and metallurgical aspects. They fabricated the component on two different materials by determining the best characteristics required. The constructed specimen was put through a tensile test and corrosion tests. The findings were compared to the casted specimens to assess the fit for alternatives. The findings have shown that WAAM is comparable to the casted product. Ahsan et al. [21] have developed the bimetallic additively manufactured structures for the SS 316 L, and low alloy carbon steel to develop the WAAM structure for two different materials by employing GMAW process. Two different low carbon steel and stainless steel zones were displayed at the welded area in the microstructure interpretation, and no defect was seen in the specified zone. In the micros-hardness testing, the hardness value was higher in the stainless steel zone due to Cr content. However, the hardness value was observed to be on the lower side for the low carbon steel zone. Thus, the tensile specimen failed from that side and that the low carbon steel side had lower strength. Lee [22] has worked on SS 316 L to evaluate the heat accumulation of the layer-by-layer deposited structure with the help of experimental and numerical studies. They observed that numerical study significantly impacted the WAAM constructed components produced from

the cold metal transfer (CMT) heat source. It was applied on 10 layered CMT-based WAAM components of SS 316 L, and then the microstructure was examined. The obtained results has shown that the grain structure of the top layer is very close to each other's, and the yield strength and hardness were predicted with an expected result. Wahsh et al. [23] identified a large-scale aluminum manufacturing method that produces layer-by-layer deposited components with the help of the pulsed arc GMAW process to deposit aluminum since it needed minimal heat input. Parts accumulated with GMAW often have no spatter, cracks, and minimal porosity, all of those frequent issues while welding aluminum alloys. After that, they conducted extensive testing to improve the input process parameters and anticipate bead shape utilizing the Robot's mode of operation. Mechanical parameters such as macrostructure and microstructure, heat contribution, and other mechanical qualities were estimated and found to be suitable. Le and Mai [24] have developed a thin-walled structure component with the 308 L stainless steel by using a GMAW process. The constructed thin walls' microstructural and mechanical properties were studied. The result has shown that columnar dendrites dominated the thin-walled 308 L components. Chen et al. [25] used metal arc additive manufacturing for the SS 316 L material to study the tensile properties at room temperature and the microstructure. From the investigations they found large number of grain structures in the center, while other grain structures are turned in the direction of a substrate surface, and generating small grain structure at the end. The phase re-dissolved in austenite throughout the heat cycles of the subsequent three deposition layers, with intermetallic phases forming at the interfaces. Subsequently, the fourth layer, phases inside austenitic dendrites, developed delicate vermicular morphologies under heat impact. GMA-AM 316 L steel has tensile qualities equivalent to wrought 316 L and meets industry standards. Another study by Ji et al. [26] used the WAAM process to produce 304 L stainless steel components. They studied mechanical properties with the help of tensile testing and microstructure by metallographic investigation. The result intemperate that when the layers are depositing to another layer, the heat input increases, the dendrites' thickness also increases, and the microstructure remains the same. Presence of dendrites the hardness will be different in the vertical and horizontal directions which may affects in mechanical properties. Their findings revelaed that presence of dendrites has changed the grain boundary strengthening influence between longitudinal and transverse directions. This in turn resulted in anisotropy of mechanical properties.

Vora et al. [27] employed GMAW-based WAAM technique for the fabrication of multi-layered structure on 2.25 Cr-1.0 Mo Steel using metal cored wire. Optimization technique was employed to produce suitable parametric settings of WAAM variables for fabrication of thin multi-layered structure. Their obtained results have shown that multi-layered structure with identical bead-on-bead deposition without any disbonding was obtained through optimized parametric settings.

The studied literature has suggested that the fabrication of multi-layered structures with minimal irregularities with favorable mechanical properties requires proper control of design variables. It is then essential to optimize the WAAM input variables for bead geometry using single-layer depositions. Therefore, the current study's aforementioned research gap has been fulfilled by examining the multi-layered structure obtained at optimized parameter settings. In the present study, GMAW-AM was used to fabricate and examine the mechanical characterizations of multi-layered structures on SS 316 L using metal wire of SS316L. The 3D structural component was made with the optimized parameters to get a defect-free and good surface finish. Optimum process variables of the WAAM process were used in the present study to fabricate the multi-layered structure. The current study concentrated on investigating the process stability and the mechanical characteristics of SS 316 L. The study focuses on the microstructure, macrostructure, tensile testing, impact testing, micro-hardness, and Scaning electron microscopy (SEM) analysis of SS 316 L. Their results are considered to be essential for future research and also for industrial applications. Authors believe that current work can solve industrial problems pertaining to the fabrication of multi-layer structures and its examination of mechanical properties.

2. Materials and methods

2.1. Experimental setup and plan

The present study employed GMAW based WAAM setup to fabricate a multi-layered structure on SS 316 L using the Miller make KEMPPI PRO MIG-530 setup. The experimental setup employed in the current study is shown in Fig. 1. Metallic wire made of SS 316 L was used to fabricate a multi-layered structure with a diameter of 1.2 mm. The chemical composition of the substrate and filler wire is represented in Table 1. The structure was constructed by shielding gas with 98% Ar and 2% O_2. During the deposition process, the substrate plate was fixed on the work-table, and a torch was used for depositing the material on the substrate plate.

In our recent study, a parametric study was conducted by employing a heat transfer search (HTS) algorithm by performing single-layer deposition [28]. The current study examined the fabricated multi-layer structure at the obtained optimized process parameters for microstructure, macrostructure, mechanical properties, and facture surface morphologies. Table 2 shows the optimal process parameters used to fabricate a multi-layered structure.

2.2. Testing and characterization

Microstructure, macrostructure, and mechanical properties of the multi-layered structure were examined at three locations at the structure's top, middle, and bottom sides. Specimens for metallographic examinations and mechanical properties were cut using wire electrical discharge machining (WEDM) processes at the top, middle, and bottom zones. The cut specimens were polished with 240, 400, 600, 800, 1000, 1200, and 2000 grit size abrasive papers and 0.5 microns alumina paste cloth paper to get the fine mirror polished surface. The components were etched for 2 min in a Hydrochloric acid (HCL): Nitric Acid (HNO_3): Glycerol solution ($C_3H_8O_3$) by the

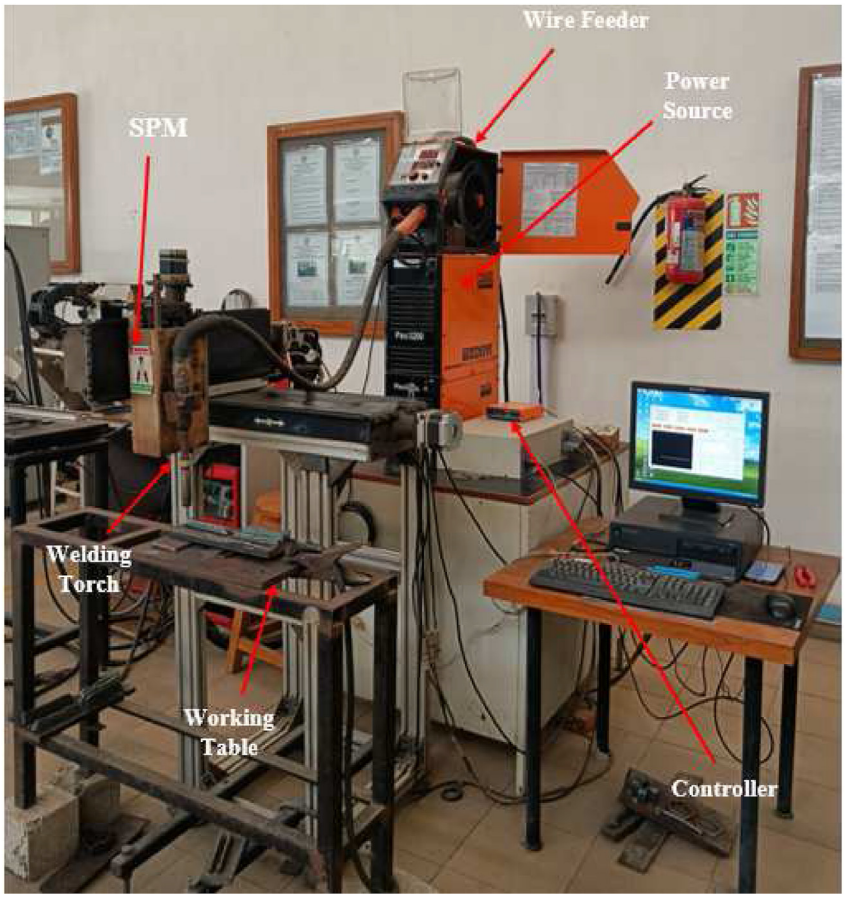


Fig. 1 – Experimental setup of WAAM process.

ratio of 40:20:40 to reveal a better microstructure, macrostructure, and surface structure. An optical microscope (Make: RAD ICAL instrument) was employed to examine microstructure, while an optimal macroscope (IMS-300-TNC) was utilized for the macrostructural analysis.

Detailed mechanical investigations of tensile testing, fractography, and microhardness were also carried out at three locations on the multi-layered structure's top, middle, and bottom sides. WEDM process was employed to cut the samples for tensile testing and impact testing. Fig. 2 shows the schematic diagram of testing locations/samples for the tensile test. A tensile specimen sample was cut as per ASTM E8M standard. M–100 universal testing machine was used for the tensile test.

The impact test was carried out using the AIT-300 Charpy impact testing machine at room temperature. The Charpy impact testing specimens were prepared with the dimensions of 55 mm × 10 mm × 3.3 mm. Fig. 2 shows the schematic diagram of testing locations/samples for impact testing.

Microhardness (MH) measurement was carried out using ESEWAY EW-150. During the Vickers hardness measurement, a 300 gf load with a dwell time of 10 s was applied for the indentations. MH was measured for the multi-layered structure's top, middle, and bottom sides. For every specimen, multiple readings were considered, and their average value was taken for the investigations.

SEM (Zeiss Ultra 55) was used to investigate the fracture surface morphologies of the tensile and impact components.

3. Results and discussion

3.1. *Fabrication of multi-layered structure*

Fig. 3 shows the Multi-layer structure fabricated at the optimal parameter setting of WAAM variables. Multi-layered structures can be seen with identical bead-on-bead deposition. To avoid generating larger residual stresses and distortion, cooling time of 15s was preferred between the consecutive layers of the multi-layer structure. Cooling time also helps in the solidification between two continuous layers. It can be observed from Fig. 3 that the obtained structure was free from

Table 1 – Chemical composition substrate and metallic wire SS 316 L.

Element	Cr	Ni	Mo	Mn	Si	C	P	S	N	Fe
Wt.%	17.09	10.61	2.38	1.17	0.59	0.013	0.011	0.011	0.09	Balance

Table 2 – Optimized variables used for the fabrication of a multi-layered structure.

Design variable	Value
Wire feed speed (WFS)	5.50 m/min
Travel speed (TS)	141 mm/min
Voltage	19 V
Arc length	3 mm
Gas flow rate	15 L/min
Weld bead length	130 mm

disbonding and also, and seamless fusion was detected among the fabricated layers. A negligible amount of metal lump was detected besides the deposition of layers. However, it can be easily eliminated during the post-processing of the structure. Therefore, the optimized process parameters successfully fabricated a thin multi-layered structure on SS 316 L substrate along with the preferred quality using GMAW based WAAM technique. Microstructure, macrostructure, and mechanical properties of the multi-layered structure were examined at three locations at the structure's top, middle, and bottom sides. Fig. 3 shows the locations of the top zone, middle zone, and bottom zone on the built structure.

3.2. *Macrostructure characterization*

A cross-sectional area was cut from the fabricated multi-layered structure for macrostructure evaluation. The component was then polished with multiple grit size abrasive (240, 400, 600, 800, 1000, 1200, and 2000) and 0.5 microns alumina paste cloth paper for obtaining the fine mirror polished surface. The component was etched for 2 min in a Hydrochloric acid (HCL): Nitric Acid (HNO_3): Glycerol solution ($C_3H_8O_3$) by the ratio of 40:20:40 to reveal a better macrostructure and surface structure. Fig. 4a–c represented the macrostructure in the built multi-layered structure's top, middle, and bottom zone, respectively. The fusion lines displayed a semi-circle morphology in these different zones and showed a transverse direction to the plate. Grains were built up at different temperatures. So, some difference was reflected in grain orientations on some levels. It suggests that the bonding between the two layers was appropriate. The macrostructures at the top, middle, and bottom have not discovered any evident interface between successive layers. This recommends a complete fusion with the absence of oxidation. The use of continuous argon shielding gas flow rate prevented oxidation [10]. The macrostructure of the extreme zones (top and bottom) was observed to be free from any unwanted geometries. The absence of porosity in the macrostructure suggests the component was free from defects. A similar macrostructure was reported by Youheng et al. [29] for bainite steel fabricated by the WAAM process. These obtained results of macrostructure have shown the suitability of the optimized process parameters of GMAW-based WAAM for fabrication of the defect-free multi-layered structure.

3.3. *Microstructure characterization*

The microstructure of the built part is as shown in Fig. 5a–c. The microstructural development in the steel is essentially a product of the cooling rates [30]. So, the microstructural observation has been carried out in 3 different zone bottom layer (first layer), the middle layer, and the top layer. The bottom zone depicting the first layer of the component was deposited on the substrate plate. As the plate and corresponding environment are at room temperature, the cooling rates experienced by these regions during the solidification are extreme [31]. Hence, the weld bead gets significantly less time for rearrangement of atoms, and a fine granular structure can be observed in Fig. 5a. Also, the liquid SS metal, on cooling

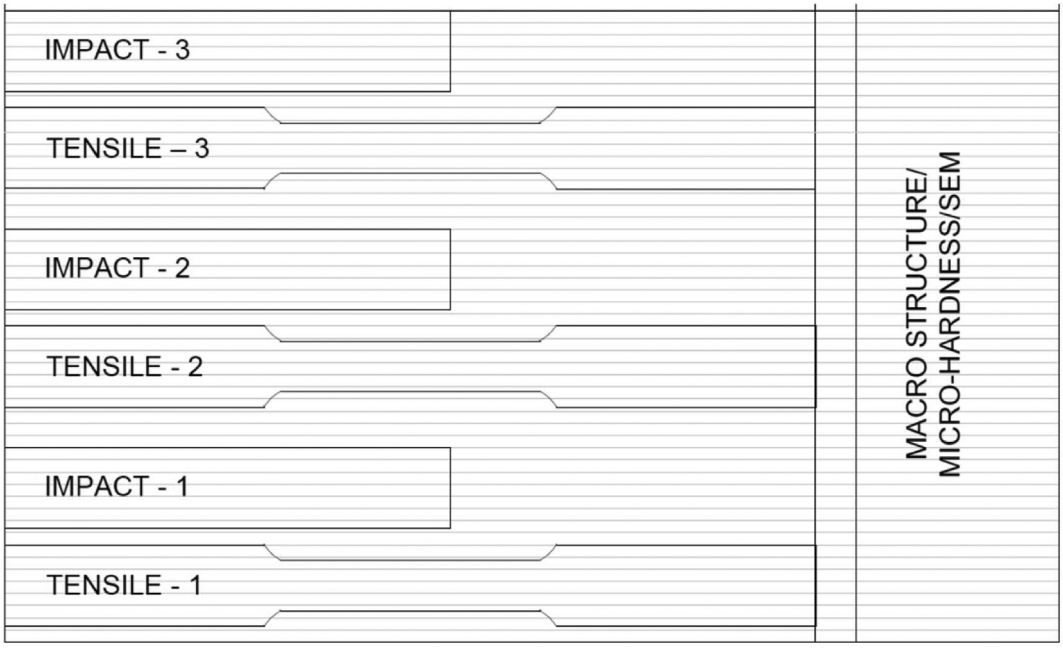


Fig. 2 – Schematic diagram representing samples for multiple characterizations.

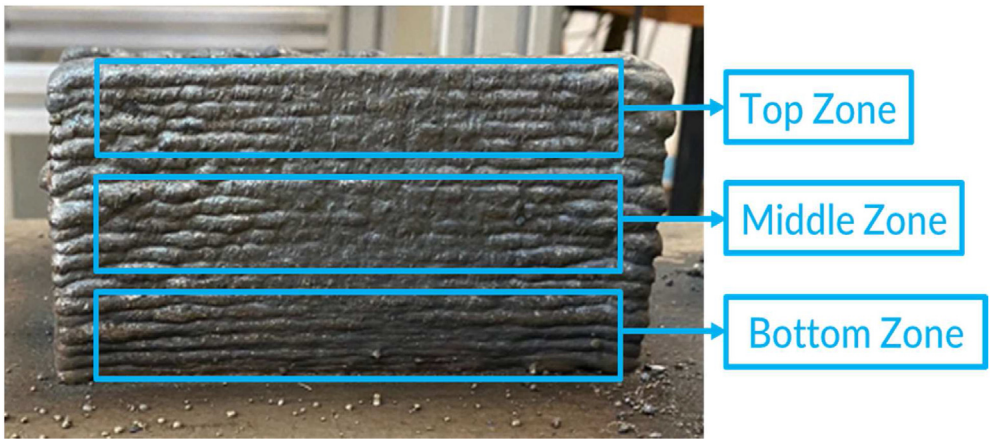


Fig. 3 – Multi-layered structure showing different zones prepared for characterizations.

and solidification, gave rise to ferrites in the shape of a skeletal. This is the inherent feature of the SS. In addition, as the weld bead is deposited using the GMAW process, a colony of columnar dendrites can be seen in the microstructure of the bottom zone, as evident from Fig. 5a. A similar microstructure was reported by Seung Hwan Lee [22] for 316 L fabricated by the CMT-based WAAM process.

The microstructure of the middle zone, as shown in Fig. 5b, depicts a different structure compared to the bottom zone. The middle zone/layers of the component are deposited on already deposited layers with higher temperatures than room temperature. Therefore, the temperature difference decreases, and hence the cooling rates also decrease. It gives rise to fairly coarser grains with vertical growth along with the residual ferrite (refer to Fig. 5b). The microstructural features of the topmost zone of the component wherein the substrate temperature are higher due to the continuously built-up of heat during previous layers are shown in Fig. 5c. A similar vertical dendritic structure and residual ferrite in skeletal shape are obtained. Identical results were reported by Gao et al. [32] for the WAAM process.

3.4. *Mechanical properties*

3.4.1. *Tensile testing*

First, mechanical properties were carried out through tensile testing at three different locations of top, middle, and bottom sides of the multi-layered structure. The tensile test specimens were prepared through a wire-cut EDM machine per the ASTM E8M. All top, middle, and bottom zone specimens showed elastic and plastic deformation before the fracture. A tensile test was also performed for wrought SS 316 L. The obtained results have demonstrated ultimate tensile strength (UTS) of 485 MPa, yield strength (YS) of 220 MPa, and percentage elongation of 45% for wrought SS316L. These values of tensile properties were compared with the results of three tensile test specimens obtained from the multi-layered structure. Table 3 summarizes the results of tensile tests for all three specimens of the multi-layered structure and wrought 316 L. Table 3 shows that the average value of top, middle, and bottom zone specimens for the UTS, YS, and elongation was obtained at 512.53 MPa, 256.57 MPa, and 49.35%, respectively. It can be concluded that experimentally

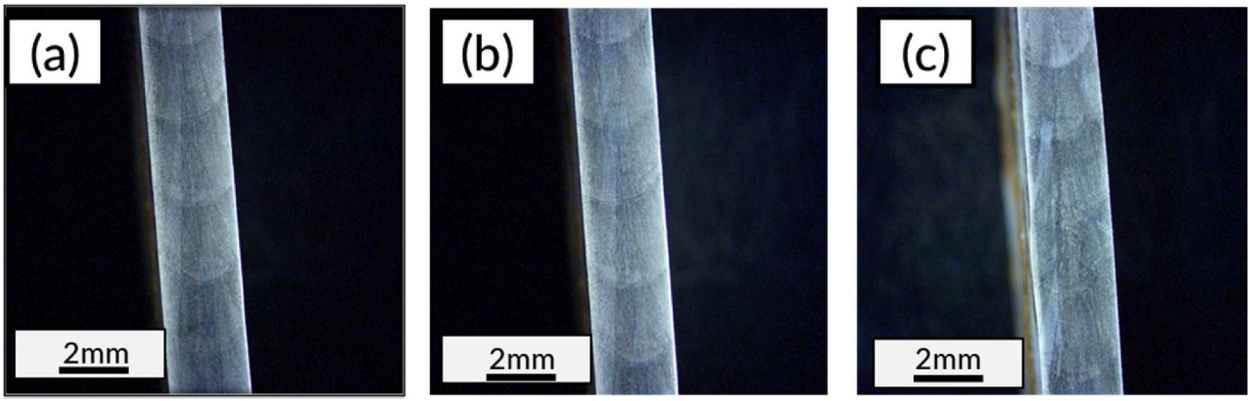


Fig. 4 – Macrostructure examination of the multi-layered structure at (a) top zone, (b) middle zone, and (c) bottom zone.

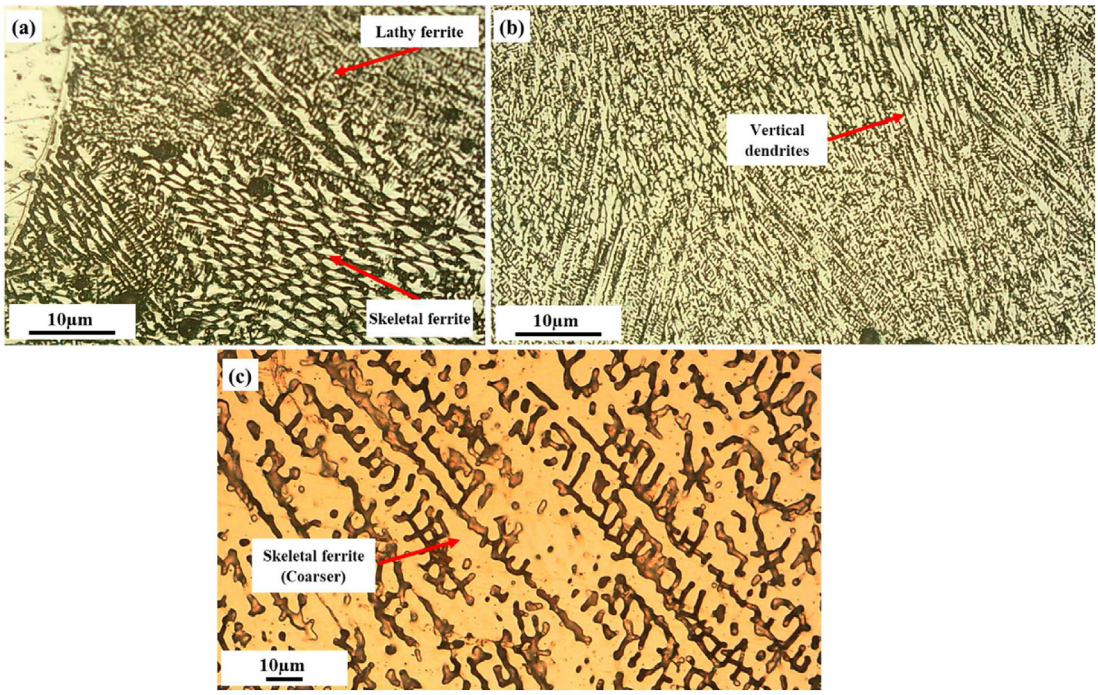


Fig. 5 – Microstructure of the multi-layered structure at (a) bottom zone, (b) middle zone, and (c) top zone.

measured UTS, YS, and Elongation of the components developed by the WAAM process fall in the range values of wrought SS 316 L. All the individual values of the top, middle and bottom zone have also shown good tensile properties compared to wrought 316 L. Thus, the GMAW-based WAAM component matches the standards for commercial use as all tensile properties validated the industry necessity for SS 316 L [20]. These obtained results are similar to Chen et al. [25]. In addition to these results, uniform tensile properties with minor variations were obtained across the top, middle, and bottom zones' multiplayer structures. Top zone values was slightly higher in comparison with the values obtained at middle, and bottom zone. As the top layer of multi-layered structure was exposed to the repeated thermal cycles due to the subsequent previous layers of formation. Due to this, a cooling rate were lesser in comparison with the bottom part. This gives rise to fairly course microstructure having slightly better YS in the top zone. However, these variations were observed to be very less and can be considered as uniform. Thus, the obtained results also proves the uniform deposition across the layers. This feature is beneficial in making additively manufactured components through GMAW-based WAAM. Suitable tensile properties of WAAM components also depicts sufficient internal superiority of produced parts for industrial applications.

Fig. 6a–b shows the fractured tensile specimen and SEM image for the fractography of the bottom zone tensile specimen. Fig. 6a shows that the fracture occurred from the gauge length after the elastic deformation. Fig. 6b depicted a large number of dimples with identical circulation on the fractured surface. It suggests a good toughness of the produced parts of the GMAW-based WAAM process [33]. A large number of dimples close to each other indicates a good ductility of the as-built structure. Similar observations were found in the top and middle zone structures as well.

3.4.2. *Impact testing*

Charpy impact test specimens were prepared per the ASTM E23-02a with 55 mm × 10 mm x 3.3 mm. For the impact test, a 2 mm V-notch of root radius 0.25 mm was cut at an angle of 45° at the center of the specimens. The test was performed at room temperature. Impact testing was performed for all top, middle, and bottom zone specimens. The obtained results have shown the values of 25.8 J, 26.2 J, and 27.8 J for top, middle, and bottom zone specimens, respectively. The average value of 26.6 J (~78 J if tested in full size as per

Table 3 – Tensile properties of GMAW-AM 316 L and wrought 316 L.

Property	UTS (MPa)			YS (MPa)			Elongation (%)		
Sample	Top	Middle	Bottom	Top	Middle	Bottom	Top	Middle	Bottom
Experimental value	520.6	512.7	504.29	268.6	249.26	251.85	48.98	49.72	49.36
Average value	512.53			256.57			49.35		
Wrought 316L values	485			220			45		

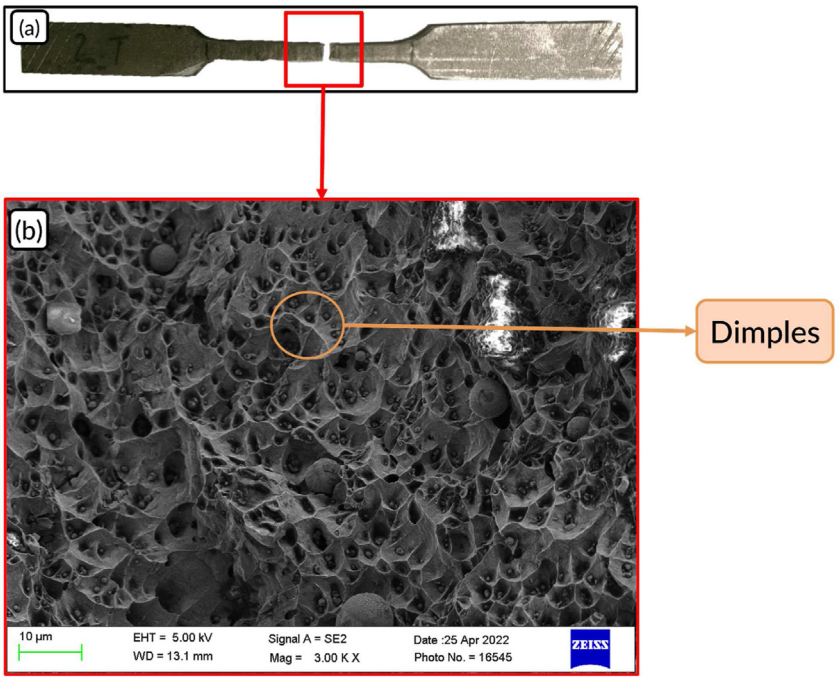


Fig. 6 – (a) Tensile specimen after fracture (b) Fractography of tensile specimen.

standard) has shown better strength than the commercially used wrought SS 316 L strength. A minor deviation was observed in the impact values. Bottom zone value was slightly higher in comparison with the values obtained at top, and middle zone. Bottom zone observed extreame cooling rate due to cold plate and atmosphere. This in turn results in slightly increase in values at the bottom zone. Therefore, the GMAW-based WAAM component matches the standards for industrial applications. Also, a uniform impact test results across the top, middle, and bottom zones' multiplayer structures suggests the uniform deposition across the layers.

Fig. 7a–b shows the fractured impact specimen and SEM image for the fractography of the bottom zone specimen. The

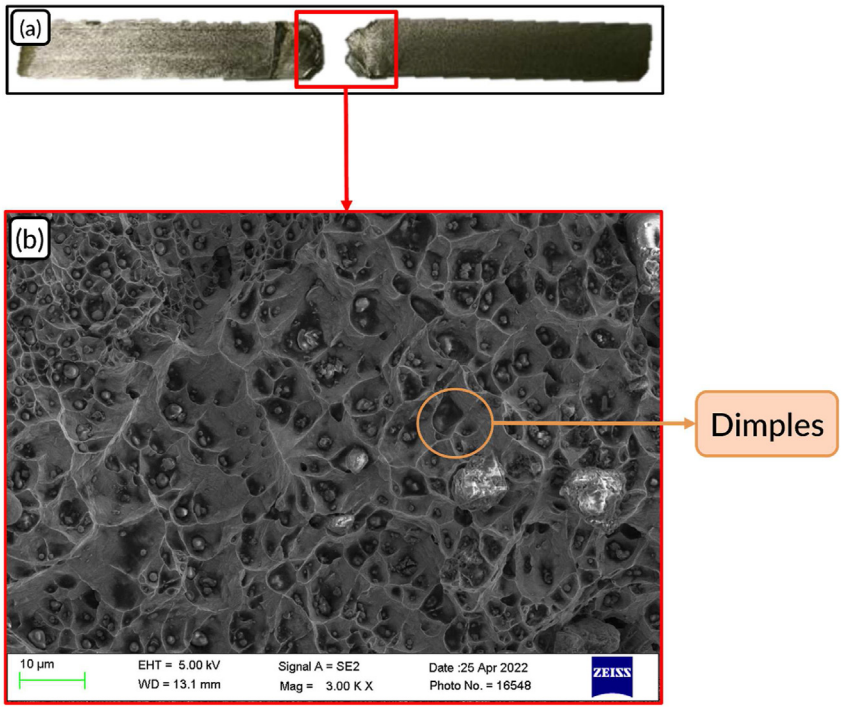


Fig. 7 – (a) Fractured specimen of impact test (b) Fractography of the impact test specimen.

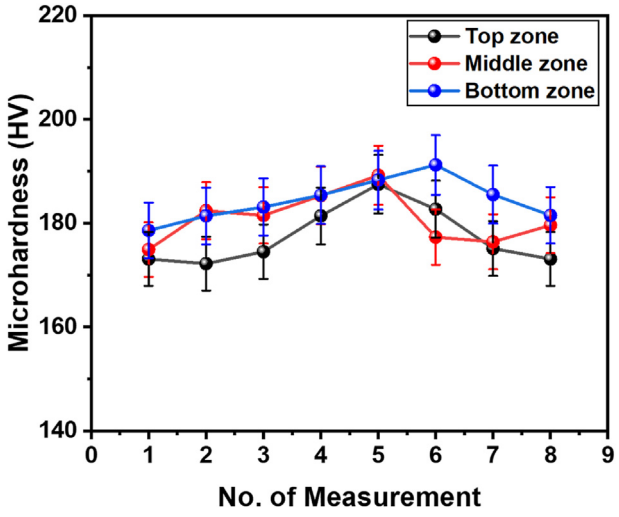


Fig. 8 – Microhardness profile of built structure.

presence of a large number of dimples from Fig. 7b suggests ductile behavior. The obtained strength from impact test results also confirms that the multi-layer deposition has excellent ductility. Additionally, without micro-cracks, micro-voids on the fracture surface depict that the specimen has good impact properties.

3.4.3. *Microhardness testing*

Fig. 8 represents the profile of the microhardness of the specimens, where the analysis was performed at 8 different locations of the multi-layered structure's top, middle, and bottom zones. Average microhardness values for top, middle and bottom zones were 177.45 H V, 180.82 H V, and 184.25 H V, respectively. Microhardness at the bottom zone was higher by a negligible amount as compared to the top and middle zone. It was due to the fact that the bottom zone depicting the first layer of the structure was to be deposited on the substrate plate and the heat-affected zone. The microhardness values showed uniform behavior across the built structure in all the zones. Thus, the uniform microhardness value suggests that it will not lead to brittle failure.

4. Conclusion

Wire-arc additive manufacturing (WAAM) process using Gas metal arc welding (GMAW) source was helpful for high deposition rate, favorable mechanical qualities, and low equipment cost. In the present study, the GMAW-based WAAM process was used to fabricate a multi-layered structure at optimized process parameters on SS 316 L using a metal wire of SS316L. Microstructure, macrostructure, and mechanical properties such as tensile testing, impact testing, microhardness, and fractography of the multi-layered structure were examined at three locations: top, middle, and bottom sides of the structure. Following are the key findings of the current study:

- The fabricated multi-layered structure was observed to be free from disbonding. Also, seamless fusion was detected among the fabricated layers. Macrostructural results have validated these findings has confirmed an appropriate bonding, complete fusion without oxidation, and defect-free structure without any unwanted geometries.
- The microstructure of the GMAW-based WAAM structure has shown a colony of columnar dendrites in the bottom zone, coarser grains with vertical growth along with the residual ferrite in the middle zone, and vertical dendritic structure with residual ferrite in skeletal shape in the top zone.
- Results of all tensile properties test for top, middle and bottom zone developed by WAAM process falls in the range values of wrought SS 316 L. UTS, YS, and elongation of wrought SS316L were found to be 485 MPa, 220 MPa, and 45%, respectively. The average value of top, middle, and bottom zone specimens for the UTS, YS, and elongation was obtained at 512.53 MPa, 256.57 MPa, and 49.35%, respectively. Thus, it can be concluded that experimentally measured UTS, YS, and Elongation of the components developed by the WAAM process fall in the range values of wrought SS 316 L.
- Average microhardness values for top, middle and bottom zones were 177.45 H V, 180.82 H V, and 184.25 H V, respectively. The microhardness values showed uniform behavior across the built structure in all the zones. It indicates that the component will not lead to brittle failure.
- Impact test results have shown the values of 25.8 J, 27.8 J, and 26.2 J for top, middle, and bottom zone specimens, respectively. The obtained average value of 26.6 J (~78 J if tested in full size as per standard) has shown better strength in comparison with the commercially used wrought SS 316 L strength. Therefore, the GMAW-based WAAM component matches the standards for industrial applications.
- SEM of fractured tensile specimens and fractures impact test specimen findings has revealed large dimples close to each other. So, obtained results clearly depicted the excellent ductility of the as-built multi-layered structure.
- Thus, obtained results have shown that built structure using the GMAW-based WAAM process matches the standards for industrial applications.
- The optimized process parameters at WFS of 5.50 m/min, TS of 141 mm/min, and voltage

Of 19 V, arc length of 3 mm, and gas flow rate of 15 L/min will be helpful for user to fabricate multi-layered structure on SS316L by using GMAW-based WAAM technique. The further studies can be attempted to analyze the multi-layered structure for advanced applications such as low-temperature strength, high-temperature sustainability etc.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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